

CSCI-665 Foundations of Algorithms

Homework 7

due Friday, December 13, 2013

Reading Chapters 34

1. Consider the following problem:

HAMILTONIAN PATH

INSTANCE: A graph G

QUESTION: Does G contain a path using all vertices in G ?

Show by direct reductions that the problems **HAMILTONIAN CYCLE** and **HAMILTONIAN PATH** are polynomially equivalent.

2. Consider the following problem:

LONGEST SIMPLE CYCLE

INSTANCE: A graph G , an integer k

QUESTION: Does G contain a simple cycle of at least k vertices ?

Show by a direct reduction that this problem is NP-complete.

3. Show that the hamiltonian-path problem can be solved in polynomial time for directed acyclic graphs by giving an efficient algorithm for this problem.

4. Suppose that an algorithm A for $x \in L \subset \Sigma^*$ runs in polynomial time in the cases YES, but it needs an exponential time in the cases NO. Argue that L can be decided by a polynomial algorithm for all strings $x \in \Sigma^*$.